

Physics-Guided and Fabrication-Aware Inverse Design of Photonic Devices Using Diffusion Models

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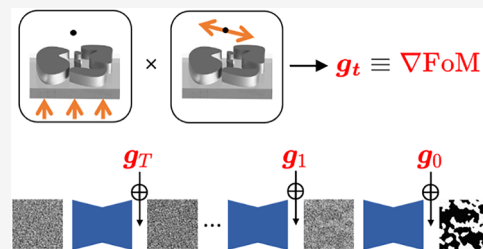
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ABSTRACT: Designing free-form photonic devices is fundamentally challenging due to the vast number of possible geometries and the complex requirements of fabrication constraints. Traditional inverse-design approaches—whether driven by human intuition, global optimization, or adjoint-based gradient methods—often involve intricate binarization and filtering steps, while recent deep-learning strategies demand prohibitively large numbers of simulations (10^5 – 10^6). To overcome these limitations, we present *AdjointDiffusion*, a physics-guided framework that integrates adjoint sensitivity gradients into the sampling process of diffusion models. *AdjointDiffusion* begins by training a diffusion network on a synthetic, fabrication-aware dataset of binary masks. During inference, we compute the adjoint gradient of a candidate structure and inject this physics-based guidance at each denoising step, steering the generative process toward high-Figure of Merit (FoM) solutions without requiring meticulous binarization or filtering. We show that our method achieves approximately 15% higher FoM at equal simulation cost compared to state-of-the-art nonlinear optimizers (e.g., Method of Moving Asymptotes (MMA), Sequential Least-Squares Quadratic Programming (SLSQP)), or requires about $3\times$ fewer simulations to reach the same FoM, all while ensuring fabrication-aware manufacturability. Compared to pure deep-learning approaches, our method requires $\sim 10^3\times$ fewer simulations. By eliminating complex binarization schedules and minimizing simulation overhead, *AdjointDiffusion* offers a simulation-efficient and fabrication-aware inverse-design algorithm with the nonconvex optimization capabilities of deep learning. Our open-source implementation is available at <https://github.com/dongjin-seo2020/AdjointDiffusion>.

KEYWORDS: inverse design, diffusion models, generative models, fabrication-aware design, adjoint optimization, photonic devices, physics-guided learning, computational photonics



1. INTRODUCTION

Photonic device design enables the control and manipulation of light in a wide array of applications, ranging from metasurfaces and photonic integrated circuits to quantum photonics and beyond. Early design efforts were guided primarily by human intuition,^{1–4} yielding simple geometries whose limited complexity often constrained device functionality. With increasing demand for multifunctional, high-performance components, free-form topologies have emerged as a powerful alternative, enabling exploration of vastly larger design spaces and unlocking superior performance.^{5–8} Photonic device design can be formulated as an optimization problem, where the objective is typically expressed as a Figure of Merit (FoM). The optimization goal is to maximize the FoM, which is often treated as a simple device efficiency but, in practice, is task-dependent (e.g., waveguide transmission, scattered-light suppression, or wavelength-selective routing). Throughout the paper, we consider FoM together with computational cost, which we regard as the two critical measures that evaluate the performance of generated designs.

To navigate the high-dimensional search spaces of photonic device designs efficiently, inverse-design techniques, partic-

ularly those based on adjoint sensitivity analysis, have gained prominence.^{9–14} By running just two electromagnetic simulations per iteration (one “forward” and one “adjoint”), these methods obtain exact gradients of FoM with respect to each design parameter. In practice, however, they demand intricate filtering and binarization routines to enforce fabrication constraints and are susceptible to convergence at suboptimal local minima.

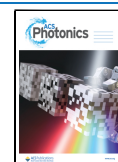
Complementary to gradient-based solvers, deep-learning frameworks,¹⁵ including generative models^{16,17} and reinforcement learning (RL) agents,^{18–20} treat inverse design from a data-driven perspective. Generative networks translate device layout design into an image generation task, while RL policies iteratively improve a design through simulated interactions. Despite their flexibility, these approaches typically require a

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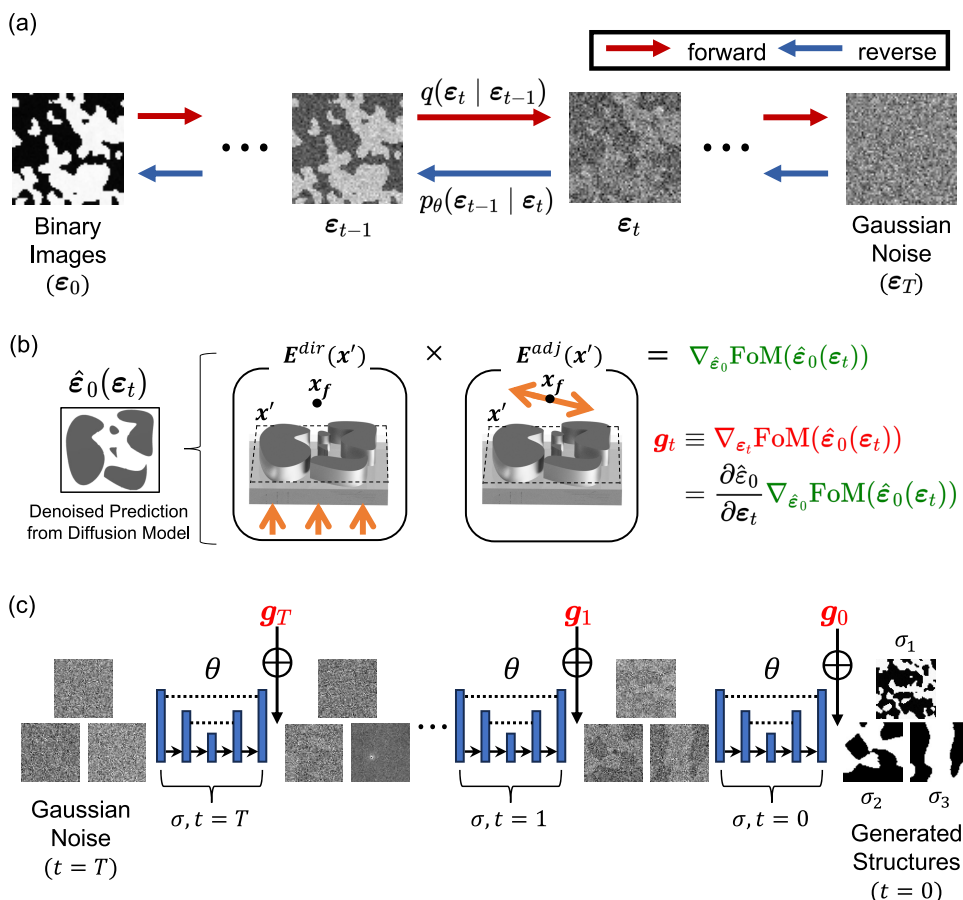


Figure 1. Illustration of the proposed framework that integrates diffusion models with adjoint optimization for photonic device design. (a) Forward and reverse-diffusion process. Starting from binary structures (ϵ_0), the forward process (red arrows) gradually adds Gaussian noise until the images become indistinguishable from pure noise (ϵ_T). The reverse process (blue arrows) denoises step-by-step using a learned model θ , reconstructing binary structures from noise. (b) Adjoint sensitivity analysis. For each denoised prediction $\hat{\epsilon}_0(\epsilon_t)$ from intermediate steps ϵ_t , the adjoint gradient $\mathbf{g}_t$ is computed as the product of direct and adjoint fields ($E^{\text{dir}}, E^{\text{adj}}$) evaluated at the design region x' . This gradient represents the sensitivity of the Figure of Merit (FoM) to the structure, obtained via the chain rule. (c) Gradient-guided reverse sampling via a U-Net-based diffusion network θ . During the reverse-diffusion process ($t = T \rightarrow 0$), the computed adjoint gradient $\mathbf{g}_t$ is added to the network output ϵ_t to steer $\hat{\epsilon}_0$ toward higher FoM while preserving fabrication-aware constraints. The diffusion schedule is controlled by the time step t , while the structural parameter σ (representing the feature length in this work) remains fixed for each dataset condition. This process continues until $t = 0$, yielding a high-FoM, fabrication-compatible structure $\hat{\epsilon}_0$.

number of simulations on the order of 10^5 – 10^6 , limiting their practicality for complex photonic problems.

In this work, we introduce *AdjointDiffusion*, a physics-guided inverse-design framework that integrates adjoint sensitivity analysis with denoising diffusion probabilistic models.^{21,22} We started this research with a simple but important question: Why do deep-learning models require so many simulations? This requirement is intrinsic to most deep-learning algorithms since they rely heavily on large-scale training data. This led us to a new idea: What if we use simulation data only during inference? Since diffusion models decouple training from sampling, they serve as a natural fit for this inference-time optimization approach.

We begin by constructing a synthetic, fabrication-aware dataset of binary structures, each smoothed via Gaussian filtering and then rebinarized, and train a conditional diffusion model to capture this fabrication-compliant prior. During inference, we generate the final structure by running the reverse-diffusion process. We then compute how the FoM changes with respect to this final design and use that gradient to guide each step of the reverse process. This guided sampling

process naturally drives the generation of high-performance, fabrication-ready layouts without the need for elaborate postprocessing or extensive hyperparameter tuning. Although diffusion models have recently been applied to photonic inverse design,^{23–26} none have embedded adjoint gradients into the sampling loop. Also, diffusion models for inverse design typically require a large amount of simulation data for training (on the order of 10^4 – 10^5 samples^{23,26}). We circumvent this limitation by training on binary data and relying on simulations during the inference phase.

We demonstrate our method on the photonic inverse-design problem of a color router for a complementary metal oxide semiconductor (CMOS) image sensor,²⁷ where a broadband incident plane wave is directed to individual photodetectors according to its wavelength (red, green, blue). Here, the FoM is evaluated as the sum of the field intensities at the detector centers across the three spectral bands. Our approach achieves comparable or better FoM than baseline algorithms while requiring only on the order of 10^2 simulations and preserving fabrication-aware manufacturability. Under the same simulation budget, it provides $\sim 15\%$ higher FoM than state-of-the-

art nonlinear optimizers such as Method of Moving Asymptotes (MMA)²⁸ and Sequential Least-Squares Quadratic Programming (SLSQP),^{29,30} or achieves the same FoM with nearly 3× fewer simulations. Compared with deep-learning-based approaches, it reduces the simulation requirement by approximately 10³ times.

We note that portions of this work were previously presented in preliminary form at conferences,^{31,32} which are nonarchival. Beyond the versions, this work investigates the nontrivial problem setup of the color-router case study and expands the analysis.

2. METHOD

Before explaining the details of our work, we provide a brief overview of diffusion models to facilitate an understanding of our key contributions. See the [Supporting Information \(S1.2\)](#) for a rigorous coverage on diffusion models.

2.1. Notation. Variables in bold denote matrices or vectors, while variables in regular font represent scalars. Note that our notation, where $\mathbf{e}$ represents the design parameter (i.e., relative permittivity), $\mathbf{x}$ represents the design space, and $\mathbf{z}$ represents random noise, differs from the convention commonly used in articles on diffusion models.

2.2. Overview of Diffusion-Based Generative Models. Diffusion models ([Figure 1a](#)) are generative frameworks that create data through an iterative refinement process.^{21,22} These models begin with a random noise sample and iteratively transform it into a structured output such as an image. Formally, let $\mathbf{e}_0 \in \mathbb{R}^d$ represent a sample from the d -dimensional data space, drawn from the data distribution p_{data} , which we denote as $\mathbf{e}_0 \sim p_{\text{data}}$. The generation process of diffusion models typically begins with random Gaussian noise $\mathbf{e}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, where $\mathbf{0}$ is the zero vector and $\mathbf{I}$ is the identity matrix, and progressively denoises it through intermediate steps $\mathbf{e}_{T-1}, \mathbf{e}_{T-2}, \dots$, until it converges to an output image $\mathbf{e}_0$, corresponding to time step 0. Here, each state $\mathbf{e}_t \in \mathbb{R}^d$ represents a d -dimensional sample at time step t where $t \in \{1, \dots, T\}$. The denoising process is performed by a neural network (typically a U-Net architecture³³) which is trained to infer a less noisy sample $\mathbf{e}_{t-1}$ from a noisier input $\mathbf{e}_t$. This denoising process is governed by a Gaussian transition model: $p_\theta(\mathbf{e}_{t-1}|\mathbf{e}_t) := \mathcal{N}(\mathbf{e}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{e}_t, t), \boldsymbol{\Sigma}_\theta(\mathbf{e}_t, t))$ where $\boldsymbol{\mu}_\theta(\mathbf{e}_t, t)$ and $\boldsymbol{\Sigma}_\theta(\mathbf{e}_t, t)$ correspond to the mean and variance governing the transition from $\mathbf{e}_t$ to $\mathbf{e}_{t-1}$.^{22,34} The mean $\boldsymbol{\mu}_\theta(\mathbf{e}_t, t)$ provides an estimate of the denoised data at step $t - 1$ by subtracting the predicted noise component from $\mathbf{e}_t$, ensuring that the sample moves closer to the original data distribution. The variance $\boldsymbol{\Sigma}_\theta(\mathbf{e}_t, t)$ accounts for the remaining uncertainty in the estimation and determines the stochasticity of the transition. This iterative refinement can be understood as a stochastic relaxation process, where the system gradually removes noise while maintaining probabilistic consistency with the original data. See [Supporting Information \(S1.2\)](#) for detailed mathematical expressions.

Concretely, at each step t , the transition from $\mathbf{e}_t$ to $\mathbf{e}_{t-1}$ is performed according to

$$\mathbf{e}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{e}_t, t) + \boldsymbol{\Sigma}_\theta^{1/2}(\mathbf{e}_t, t)\mathbf{z}, \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \quad (1)$$

Given a well-trained diffusion model (i.e., trained until convergence), it is theoretically guaranteed that iterating this process ([eq 1](#)) from $t = T$ down to $t = 1$ reconstructs samples

that are statistically close to the original data distribution, i.e., $p_\theta \approx p_{\text{data}}$.

2.2.1. Conditional Generation via Guidance. A key advantage of diffusion-based samplers is that they admit interventions in the iterative denoising process to implement conditional generation.³⁵ Specifically, given a differentiable target utility function $f(\mathbf{e}_t, t, \mathbf{c})$ that may depend on an arbitrary condition $\mathbf{c}$, one can incorporate its gradient into the update [eq 1](#) to yield a guided sampler. We denote this modified inference equation as ([eq 2](#)):

$$\begin{aligned} \mathbf{e}_{t-1} &= \boldsymbol{\mu}_\theta(\mathbf{e}_t, t) + \boldsymbol{\Sigma}_\theta^{1/2}(\mathbf{e}_t, t)\mathbf{z} + \eta_t \nabla_{\mathbf{e}_t} f(\mathbf{e}_t, t, \mathbf{c}), \\ \mathbf{z} &\sim \mathcal{N}(\mathbf{0}, \mathbf{I}) \end{aligned} \quad (2)$$

[Equation 2](#) consists of three components. The first term, $\boldsymbol{\mu}_\theta(\mathbf{e}_t, t)$, captures the mean of the data distribution at each step. The second term, $\boldsymbol{\Sigma}_\theta^{1/2}(\mathbf{e}_t, t)\mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, reflects the learned variance and introduces stochasticity, which promotes sample diversity and enables escape from local optima. The third term, $\eta_t \nabla_{\mathbf{e}_t} f(\mathbf{e}_t, t, \mathbf{c})$, adds a guidance term that steers the generation toward desired conditions or objectives, where η_t corresponds to the strength of guidance term which may be scheduled over inference step t . An example of such guidance is classifier guidance,³⁵ where the log-likelihood of a noise-aware class predictor is incorporated: $f(\mathbf{e}_t, t, y) := \log p_\phi(y|\mathbf{e}_t)$. Here, ϕ represents the parameters of the classifier, and y denotes the selected condition for the generated output.

Although effective, many guided samplers, including classifier guidance, require potentially expensive additional resources such as data collection and model training. This requirement limits the applicability of these techniques in challenging scenarios, such as our focused inverse-design problem, where obtaining a large volume of high-quality structural data is prohibitive. In the next section, we develop a data-free framework for inverse design by intelligently incorporating adjoint sensitivity analysis into the concept of guided sampling, which offers significant performance improvements over traditional inverse-design solvers with minimal implementation complexities.

2.3. AdjointDiffusion: Physics-Guided Diffusion Models for Inverse Design. We propose our physics-guided diffusion model algorithm named *AdjointDiffusion*, which combines adjoint sensitivity analysis with a generation pipeline of diffusion models. Our approach consists of four distinct stages: (i) dataset construction, (ii) diffusion model training, (iii) structure generation, and (iv) postprocessing.

2.3.1. Dataset Construction. We first prepare training data for constructing a diffusion model (i.e., a denoising network p_θ) specifically designed for our inverse-design framework. To generate structured training data, we first define a design parameter ranging from 0 to 1, where 0 means a material with lower permittivity and 1 means a material with higher permittivity. Then, we start by creating 30,000 random binary (design parameter with 0 or 1) data with a resolution of 64×64 as shown in [Figure 1a](#). Generating 30,000 binary samples of 64×64 resolution is computationally inexpensive, taking only 16.93 ± 0.13 s with a simple Python script.

To incorporate fabrication constraints into the dataset, we apply Gaussian filters to the randomly generated structures, which smooth the sharp edges of the material distribution in space, resulting in a design parameter distribution with diverse

density values ranging from 0 to 1.³⁶ The dataset is evenly divided into three subsets of 10,000 images, each processed with a Gaussian filter of standard deviation 2, 5, or 8. The filtered structures are then binarized using a 0.5 threshold, enforcing two-material (0 and 1) composition. With this approach, we enable the conditional generation of structures within a single network, allowing it to accommodate multiple fabrication constraints while being trained only once. Also, since our algorithm can inherently work with any dataset by simply training the diffusion model on it, one can apply their datasets following different designs and applications.

2.3.2. Diffusion Model Training. With the generated training dataset, we subsequently train a diffusion model p_θ following the methodology proposed by Dhariwal and Nichol.³⁵ Specifically, we use 1000 timesteps (i.e., $T = 1000$) with the cosine noise scheduling.³⁷ For the training objective, we adopt the hybrid loss L_{hybrid} introduced by Nichol and Dhariwal.³⁷ For each training sample, the associated fabrication constraint (i.e., the Gaussian filter's standard deviation) is incorporated as a conditioning parameter σ , where the associated Gaussian transition is then expressible as $p_\theta(\mathbf{e}_{t-1}|\mathbf{e}_t, \sigma) := \mathcal{N}(\mathbf{e}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{e}_t, t, \sigma), \boldsymbol{\Sigma}_\theta(\mathbf{e}_t, t, \sigma))$. This training enables the diffusion model to generate binary data that meet the specified fabrication conditions σ . See the [Supporting Information \(S1.2 and S2\)](#) for details on the training process.

2.3.3. Structure Generation. The third stage, which forms the core of our framework, involves generating high-performance photonic structures with enhanced FoM. One straightforward approach to achieve this is to select a desired feature size σ and generate binary structures by applying the following equation (iteratively from $t = T$ down to $t = 1$):

$$\mathbf{e}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{e}_t, t, \sigma) + \boldsymbol{\Sigma}_\theta^{1/2}(\mathbf{e}_t, t, \sigma)\mathbf{z} \quad (3)$$

However, as demonstrated in our results (see [Supporting Information, S3](#)), this approach does not yield structures with high-FoM values. An alternative is to use a diffusion model trained on high-FoM structural data for sampling, but as previously mentioned, this approach is resource-intensive due to the difficulty of obtaining such high-quality data.

We alter this process by leveraging the principle of guided sampling in diffusion models. Specifically, we intervene in the sampling process (eq 3) and incorporate the FoM-maximizing direction, calculated using the adjoint sensitivity analysis, to guide the generation process toward high-FoM regions of the data space.

To achieve this, we first compute the posterior mean $\hat{\mathbf{e}}_0(\mathbf{e}_t, \sigma) := \mathbb{E}[\mathbf{e}_0|\mathbf{e}_t, \sigma]$ by employing the pretrained diffusion model and Tweedie's formula³⁸ (see [Supporting Information S1.2](#) for details on the formula). Next, we calculate the adjoint gradient of the predicted structure, $\nabla_{\mathbf{e}_0}\text{FoM}(\hat{\mathbf{e}}_0(\mathbf{e}_t, \sigma))$, using direct and adjoint simulations^{11,12} (see [Figure 1b](#)). By computing the Jacobian $\partial\hat{\mathbf{e}}_0/\partial\mathbf{e}_t$ and taking the inner product with the adjoint gradient $\nabla_{\mathbf{e}_0}\text{FoM}(\hat{\mathbf{e}}_0(\mathbf{e}_t, \sigma))$, we obtain the proposed guidance term, which directly follows from the chain rule:

$$\nabla_{\mathbf{e}_t}\text{FoM}(\hat{\mathbf{e}}_0(\mathbf{e}_t, \sigma)) = \frac{\partial\hat{\mathbf{e}}_0}{\partial\mathbf{e}_t}\nabla_{\mathbf{e}_0}\text{FoM}(\hat{\mathbf{e}}_0(\mathbf{e}_t, \sigma))$$

See [Figure 1b](#) for illustration. The proposed sampler, which is illustrated in [Figure 1c](#), is a U-Net-based diffusion network θ that captures spatial features across multiple scales, providing

stable denoising and smooth integration of adjoint-guided optimization. This network integrates the calculated guidance into the standard sampling process (eq 3), as formally expressed by

$$\mathbf{e}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{e}_t, t, \sigma) + \boldsymbol{\Sigma}_\theta^{1/2}(\mathbf{e}_t, t, \sigma)\mathbf{z} + \eta_t\nabla_{\mathbf{e}_t}\text{FoM}(\hat{\mathbf{e}}_0(\mathbf{e}_t, \sigma)) \quad (4)$$

where η_t represents the strength of guidance term possibly scheduled over time t . Note that the guidance term in eq 4 aims to optimize intermediate structures $\mathbf{e}_t$ by incorporating the gradient $\nabla_{\mathbf{e}_t}\text{FoM}(\hat{\mathbf{e}}_0(\mathbf{e}_t, \sigma))$ instead of directly using $\nabla_{\mathbf{e}_t}\text{FoM}(\mathbf{e}_t)$. This formulation allows gradients to be computed from the finally denoised structure, which provides clearer directions for improving the device performance. Such guidance benefits from the manifold-constrained behavior observed in recent studies,^{39–42} leading to better stability and sample quality.

Specifically, the Jacobian $\partial\hat{\mathbf{e}}_0/\partial\mathbf{e}_t$, which appears in the calculation of the guidance term, can be interpreted as a projection onto the tangent space of the data manifold $\mathcal{M}$ at $\hat{\mathbf{e}}_0$, denoted $\mathcal{T}_{\hat{\mathbf{e}}_0}\mathcal{M}$. This makes the guidance term point in the tangent direction relative to $\mathcal{M}$, thereby constraining the updated structures to reside on the data manifold. The manifold-constrained property offers significant empirical benefits. For instance, it naturally encourages the binarization of generated structures during the generation process. This is a crucial advantage since it decreases the need for the binarization process, which requires tuning numerous hyperparameters. See the [Supporting Information \(S5\)](#) for experimental results demonstrating this advantage.

2.3.4. Postprocessing. To ensure that the final output $\mathbf{e}_0$ corresponds to a fabricable photonic configuration, we apply a simple postprocessing pipeline that transforms $\mathbf{e}_0$ into a binary image suitable for fabrication. This postprocessing consists of two stages: (1) final binarization and (2) island deletion. First, since our algorithm does not use conic filters,⁴³ which act as a blurring kernel to smooth the design variables, there is a possibility that a pixel with an adjoint gradient value has an opposite sign from nearby pixels. This issue can degrade the fabricability due to the low minimum feature size (MFS). The problem is also observed in nonlinear algorithms typically used in adjoint-based optimizations, such as those provided in NLOpt.⁴⁴ To address this, we apply postprocessing after binarization across all algorithms.

We highlight several advantages of the proposed inverse-design framework. First, it operates in a data-free manner, allowing the synthesis of high-quality structures without requiring real-world data. Second, our framework exhibits minimal reliance on hyperparameters. By employing a diffusion model trained on binary data, the sampling process directly generates binary structures, eliminating the need for complex binarization schedules and extensive hyperparameter tuning. Finally, our algorithm demonstrates high computational efficiency, requiring significantly fewer adjoint simulations ($\sim 2 \times 10^2$) compared with existing deep-learning methods.

3. RESULTS AND DISCUSSION

In this work, we take the FoM to be the primary optimization objective but always evaluate it after fabrication-aware processing, so that it represents the performance of realizable devices. Alongside FoM, we also consider the computational

cost of an algorithm as a critical metric of practicality in photonic inverse design. Accordingly, throughout this section, performance is assessed in terms of both FoM and computational cost. We evaluate the performance of *AdjointDiffusion* in comparison to conventional nonlinear optimization algorithms widely used in adjoint optimizations, such as MMA²⁸ and SLSQP;^{29,30} details of these algorithms are provided in Supporting Information (S1.1). Due to the stochastic nature of our algorithm, we perform three independent optimization runs using different random seeds to demonstrate robustness and reproducibility; each run generates different structures, offering diverse design options.

3.1. Preliminaries. **3.1.1. Number of Simulations.** From the viewpoint of computational efficiency, the number of simulations required for the algorithm is an important factor. Especially in large-scale tasks, such as large-area or three-dimensional (3D) simulations, the simulation step typically dominates the overall computational cost. This makes the number of simulations a reliable proxy for the computational cost. Adjoint sensitivity analysis performs two (direct and adjoint) simulations to calculate the adjoint gradient of the generated structure, resulting in an overall simulation requirement that is twice the number of optimization steps. We note that the reverse process of *AdjointDiffusion* can take any step numbers,^{35,37} and we set the step numbers as 60, 80, 100, 120, and 140. For a fair comparison, baseline algorithms based on adjoint sensitivity analysis were evaluated by using the same step numbers as our algorithm.

3.1.2. Structural Characteristics. The feature length is set to three distinct values: 0.224, 0.562, and 0.895 μm , corresponding to Gaussian filter standard deviations of $\sigma = 2, 5$, and 8, respectively. The derivation of these values is detailed in Supporting Information (S4). Additionally, we characterize the generated structures by quantifying the number of “islands,” which represent clusters of high-refractive-index regions and evaluating their Minimum Feature Size (MFS). A lower number of islands is preferred for fabrication, as it reduces structural fragmentation, while a larger MFS is desirable to ensure manufacturability.

3.1.3. Pre/Postprocessing. To ensure a fair comparison, identical structural constraints (Gaussian filtering for the dataset of *AdjointDiffusion*, conic filter for baseline algorithms) are applied across all methods as explained in Supporting Information (S4). Postprocessing, which consists of threshold-based binarization at 0.5 and island deletion, is also applied identically for all methods.

3.2. Problem Setup: Color-Routing-Based CMOS Image Sensor. Our problem setup, illustrated in Figure 2, focuses on a color-routing-based CMOS image sensor (CIS) operating in the visible spectrum. The simulation employs Bloch boundary conditions on the left and right boundaries and a perfectly matched layer (PML) at the top, following the configuration of Lee et al.²⁷ Specifically, we aim to route red, green, and blue (RGB) light to different spatial locations to enable color separation directly at the pixel level. To achieve this, we introduce red (650 nm), green (550 nm), and blue (450 nm) light as spatially overlapping, vertically incident beams from the top of the structure. Each color channel is represented by a Gaussian-shaped pulse centered at its respective wavelength with a 40% relative bandwidth to ensure sufficient spectral coverage. The incident beam uniformly spans the horizontal extent of the design region and is polarized along the out-of-plane direction, stimulating in-plane

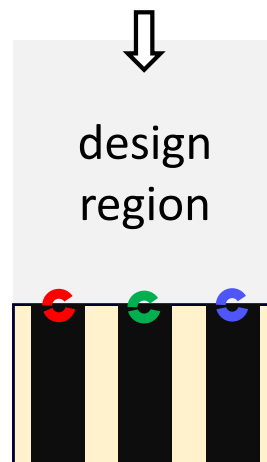


Figure 2. Problem setup: CMOS image sensor (CIS) configuration with a 64×64 pixel design region. The white arrow indicates the broadband (white light) source. Bloch boundary conditions are applied to the left and right and a PML to the top boundary. Colored circles indicate the monitor locations for red, green, and blue (R, G, B).

electric field responses. For this design, we employ silicon nitride (Si_3N_4) with $\epsilon_r = 4$ and silicon dioxide (SiO_2) with $\epsilon_r = 2.1$ as the available materials. The design region consists of a 64×64 pixel grid. By tailoring the material distribution, the device directs each wavelength band (red, green, and blue) to a separate output waveguide, thereby achieving wavelength-selective routing. By directing incident light to the appropriate subpixel, depending on its wavelength, the routing strategy mitigates the efficiency loss inherent to conventional filter-based CMOS image sensors.

Initially, the FoM for each wavelength is computed via direct simulation, where a Gaussian source is incident from the top of the design region. The FoM is then formulated as

$$\text{FoM} = \sum_{\lambda \in \Lambda_R} |\mathbf{E}(\mathbf{x}_R, \lambda)|^2 + \sum_{\lambda \in \Lambda_G} |\mathbf{E}(\mathbf{x}_G, \lambda)|^2 + \sum_{\lambda \in \Lambda_B} |\mathbf{E}(\mathbf{x}_B, \lambda)|^2 \quad (5)$$

where $\mathbf{x}_R$, $\mathbf{x}_G$, and $\mathbf{x}_B$ denote the central positions of the photodetectors in each subpixel, while $\mathbf{E}$ represents the electric field. The visible spectrum is divided into three wavelength bands: 400–500 nm for blue (Λ_B), 500–600 nm for green (Λ_G), and 600–700 nm for red (Λ_R). Within each spectral band, field intensities are sampled at discrete wavelengths, and the total efficiency is computed by summing the intensity contributions across these wavelengths. During optimization, the material values in the design region are constrained to lie within the ranges of SiO_2 (represented as 0) and Si_3N_4 (represented as 1), ensuring that intermediate permittivity values do not exceed physically meaningful limits.

3.3. Optimization Performance across Algorithms. We present the optimization process and results of our algorithms compared with the baseline algorithms for two target problems. Figure 3a illustrates the optimization process of multiple algorithms optimizing the structure of CMOS color routers. To ensure a fair comparison, all methods were run for the same number of iterations (100), with a total of 200 simulations performed for each approach. While all methods utilize adjoining gradients, their application varies: MMA and

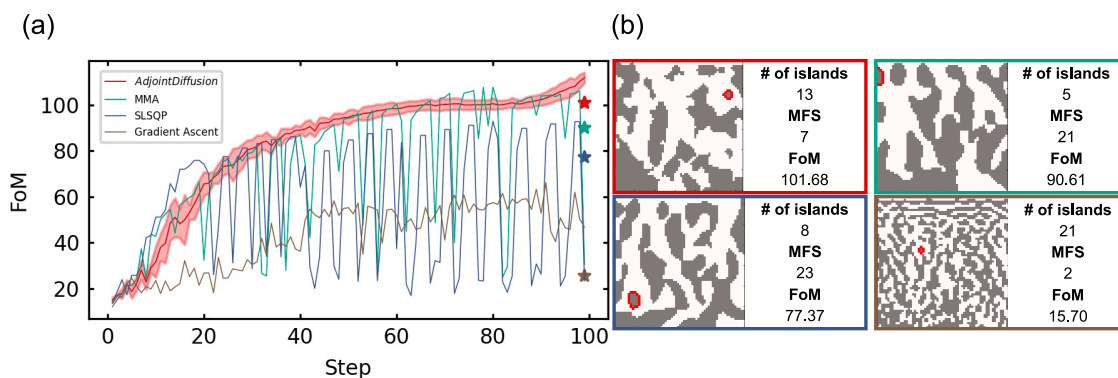


Figure 3. Optimization trajectories and resulting structures of CMOS color-router designs under fixed fabrication constraints (feature length = $0.224 \mu\text{m}$). (a) FoM as a function of the optimization step for *AdjointDiffusion* and baseline algorithms. The total optimization step for each algorithm is set to 100. For *AdjointDiffusion*, the red line denotes the mean FoM over three random seeds with shaded regions showing the standard deviation. The FoM after postprocessing is indicated by a star symbol. All methods use the same total number of simulations ($2 \times \text{step}$) for fair comparison. (b) Structures generated by each method, annotated with three key characteristics: number of islands (contiguous high-index regions), minimum feature size (MFS), and the final FoM. The smallest island that determines the MFS is highlighted in red for each case.

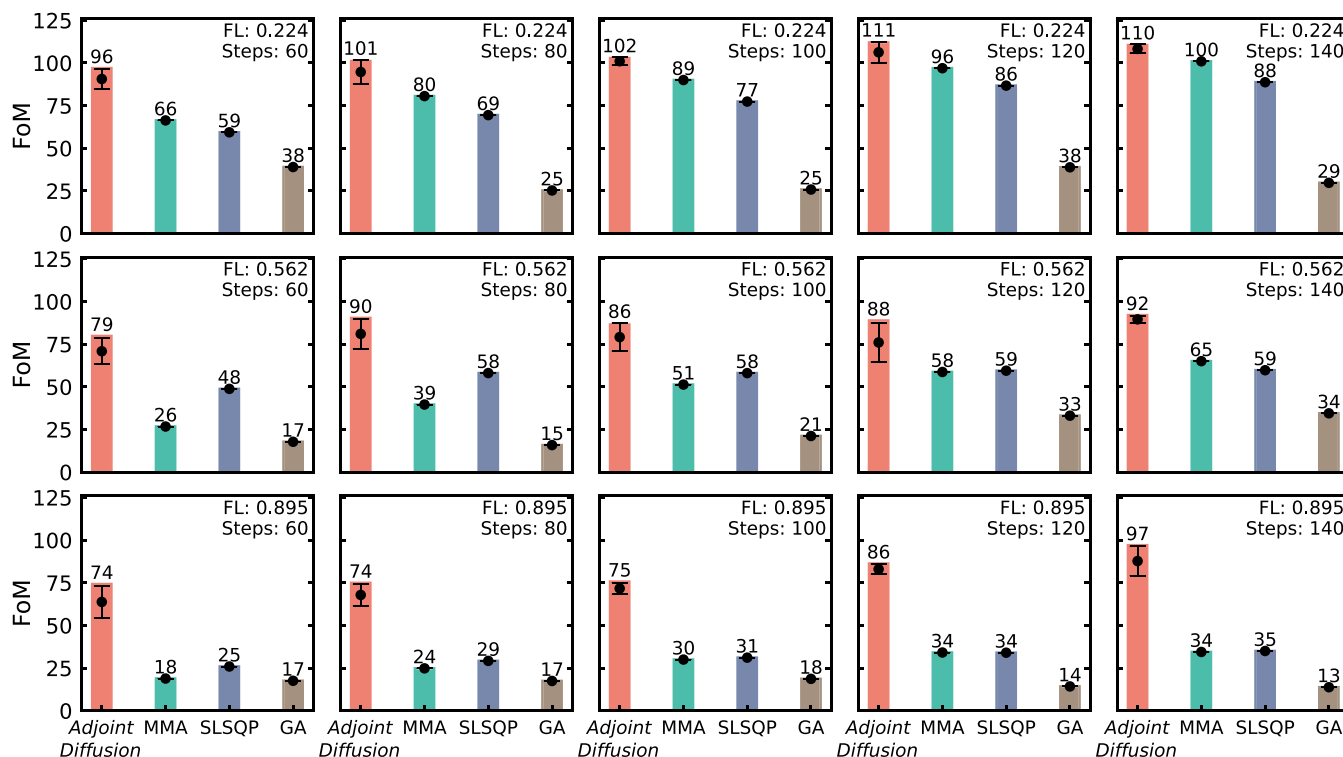


Figure 4. Maximum FoM (summed light intensity at the detectors) of optimized color-router structures across different feature lengths (FL) and optimization steps. Three FL values [μm] are considered: 0.224 (top row), 0.562 (middle row), and 0.895 (bottom row). *AdjointDiffusion* results are averaged over three random seeds with error bars showing the standard deviation, while baseline methods are deterministic. All methods use the same total number of simulations ($2 \times \text{steps}$) for a fair comparison. GA denotes the standard Gradient Ascent algorithm.

SLSQP employ nonlinear optimization techniques, *AdjointDiffusion* applies gradients within the reverse process of a diffusion model, and Gradient Ascent (GA) simply adds the gradient update. Note that baseline algorithms (MMA, SLSQP, and Gradient Ascent) require a meticulous binarization process and conic filters.⁴³ The star-shaped markers represent the efficiency after a direct threshold-based binarization at 0.5. For *AdjointDiffusion*, the red star denotes the average performance across three different random seeds. The results demonstrate that our method not only discovers optimal structures but also exhibits robustness to the binarization process. This robustness stems from the fact that our algorithm is trained on binary

data, inherently guiding the generated structure toward binarity without requiring explicit binarization tuning. In contrast, baseline methods rely on scheduled binarization controlled by multiple hyperparameters, making them more sensitive to the choice of binarization strategy. Also, although baseline algorithms can be followed by postprocessing of binarization and island deletion, we observe that the final designs often become brittle after this process, leading to degraded performance compared to *AdjointDiffusion*. Figure 3b presents the final structures obtained by each optimization algorithm along with key structural characteristics such as the number of isolated high-refractive-index regions (islands) and

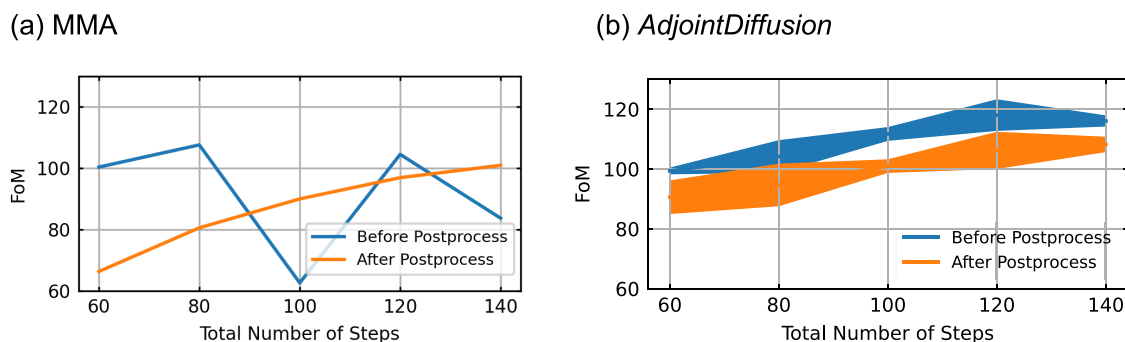


Figure 5. Comparison of FoM before and after postprocessing for deterministic and stochastic optimization methods with the feature length set as $0.224\ \mu\text{m}$. (a) MMA (deterministic adjoint optimization): Each line represents FoM before (blue) and after (orange) postprocessing over various numbers of optimization steps. Due to its deterministic nature, MMA exhibits significant FoM degradation at low step numbers after binarization and island deletion; the performance can be recovered only by substantially increasing the number of optimization steps, which considerably raises the simulation cost. (b) *AdjointDiffusion* (stochastic optimization): The shaded regions indicate the mean $\pm$ standard deviation of FoM before (blue) and after (orange) postprocessing. The stochasticity of *AdjointDiffusion* enables broader exploration of the design space, leading to higher FoM and more robust postprocessed performance with minimal degradation and low variance across random trials.

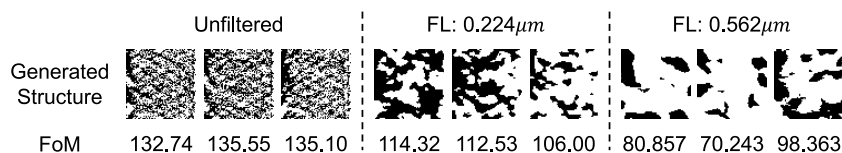


Figure 6. Representative generated structures and corresponding FoMs illustrating the trade-off between maximum achievable performance and fabrication-aware robustness. “Unfiltered” refers to the dataset where no Gaussian filter was applied, which yields the highest FoM but produces dense, fabrication-unfriendly patterns. Applying Gaussian filters with feature lengths of 0.224 and $0.562\ \mu\text{m}$ progressively enforces larger feature sizes, improving fabricability at the cost of a reasonable reduction in FoM.

the Minimum Feature Size (MFS). The red boundaries highlight the smallest MFS regions in each structure. While some algorithms may show slight advantages in structural characteristics, the overall differences are minor, except for Gradient Ascent, which produces highly fragmented and impractical designs. *AdjointDiffusion* maintains a reasonable balance between the number of islands and the MFS, producing a manufacturable structure without excessive fragmentation.

3.3.1. Computational Cost. A key advantage of *AdjointDiffusion* is its favorable computational cost compared to deep-learning-based approaches. Importantly, our method uses simulations only during inference and does not require retraining for new objectives, unlike conventional deep-learning-based methods that require large amounts of simulation data and retraining for each new task. As a result, the total computational cost of *AdjointDiffusion*, including training, remains consistently lower than that of standard deep-learning algorithms. However, we note that compared to the adjoint method, our approach introduces an intentional trade-off: a slightly higher per-step computational cost due to added diffusion steps, in exchange for improved device performance and robust, fabrication-aware designs under limited simulation budgets. In practice, the total number of steps required by the adjoining method is typically much larger to achieve high performance, leading to even higher overall cost, especially in expensive settings such as large-area or 3D simulations. Moreover, the training cost of our algorithm is negligible in practice, as the trained model can be reused across multiple design tasks without retraining. For a detailed comparison and analysis of computational cost, see [Supporting Information \(S6\)](#).

3.3.2. FoM across Diverse Conditions. In [Figure 4](#), we compare the optimization results of *AdjointDiffusion* with multiple baseline algorithms across a range of design conditions (feature lengths (FL) of 0.224 , 0.562 , and $0.895\ \mu\text{m}$) and step numbers (60 , 80 , 100 , 120 , and 140). The results show that *AdjointDiffusion* consistently outperforms nonlinear optimization methods and the standard Gradient Ascent (GA) algorithm under all tested scenarios. The error bars in the red graphs indicate standard deviations due to stochasticity, which remain relatively small and do not significantly impact the overall performance trends. Note that Gradient-Ascent-based baseline algorithms such as MMA, SLSQP, and standard Gradient Ascent are deterministic and, therefore, do not exhibit variance in their results.

3.4. Analysis. **3.4.1. Robustness to Postprocessing and Better Search from Stochasticity.** We analyze the degradation of the FoM after postprocessing steps, including binarization and island deletion. As shown in [Figure 5](#), the performance advantage of *AdjointDiffusion* does not originate merely from postprocessing filtering, but from its operation within a fabrication-aware manifold. In [Figure 5a](#), using MMA as a representative deterministic optimizer, we observe that although it can achieve a high FoM during the optimization process, it is particularly vulnerable to postprocessing steps such as binarization and island deletion when the number of optimization steps is small. In contrast, the stochastic search strategy employed by *AdjointDiffusion* ([Figure 5b](#)) explores a broader design space and generally achieves higher FoM values after postprocessing. Furthermore, the relatively small standard deviation across stochastic trials indicates that *AdjointDiffusion* consistently discovers robust, high-performing structures rather than relies on random chance. While deterministic baseline algorithms can partly alleviate this degradation by increasing

the number of optimization steps, doing so substantially increases the overall simulation cost. The results demonstrate the simulation efficiency and robustness advantages of the stochasticity in the *AdjointDiffusion* framework.

3.4.2. Trade-off between Maximum Achievable FoM and Fabrication-Aware Robustness. We quantify the trade-off between maximum achievable FoM and fabrication-aware robustness by varying the Gaussian filter strength applied to the random binary data used for diffusion model training. Representative generated structures and their FoM values (three random seeds per condition) are shown in Figure 6. Unfiltered training achieves the highest FoM (≈ 130.00) but generates dense, island-rich patterns that are difficult to fabricate. In contrast, enforcing a larger feature length (e.g., $0.562\ \mu\text{m}$ in the proposed conditional diffusion model) produces smoother, fabrication-friendly structures with only a modest reduction in FoM. This clearly demonstrates a controllable trade-off: stronger filtering enhances fabrication robustness at the cost of slightly reduced performance. Overall, the result confirms that *AdjointDiffusion* effectively balances performance and manufacturability in a controllable and physically interpretable manner.

4. CONCLUSIONS

We have presented *AdjointDiffusion*, a physics-guided inverse-design framework that integrates adjoint sensitivity analysis with denoising diffusion models to rapidly generate high-performance, fabrication-compliant photonic structures. By training on a synthetic binary dataset and injecting adjoint gradients directly into the reverse-diffusion process, our approach obviates complex binarization steps and reduces the required simulation budget to the order of 10^2 runs. At equal simulation cost, *AdjointDiffusion* achieves $\sim 15\%$ higher FoM than state-of-the-art nonlinear optimizers (MMA, SLSQP), or attains the same FoM with $\sim 3\times$ fewer simulations, while requiring $\sim 10^3\times$ fewer simulations than deep-learning-based methods. Beyond computational cost, our framework explicitly incorporates fabrication-aware constraints, ensuring robust device performance after postprocessing and enabling a controllable trade-off between maximum FoM and manufacturability. We have validated our algorithm on a CMOS color-router design task, where it outperformed conventional adjoint-based optimizers in FoM while showing comparable manufacturability. These results establish our method as a simulation-efficient and generalizable framework for photonic inverse design with potential applicability to next-generation photonic device development under fabrication-aware constraints.

5. LIMITATIONS AND FURTHER WORKS

Though we have limited our target problem to photonic inverse design, our algorithm is not restricted to photonics. Since adjoint sensitivity analysis is a broadly applicable technique, our method can be extended to other domains such as mechanics⁴⁵ and fluid dynamics.⁴⁶ Also, while the application in this paper is limited to different feature sizes of the input structure, the applicability of our algorithm is unlimited; any structural constraints can be incorporated into the generation process by training on an image dataset of constrained structures. For example, our method can generate designs that adhere to sophisticated fabrication constraints, such as in Schubert et al.⁴⁷ On top of that, the current

application of our work is limited to two-dimensional structures. However, with advancements in 3D simulations and the development of 3D diffusion models, our approach can be extended to three-dimensional design optimization. This expansion would allow for the optimization of volumetric structures and will enable applications such as multilayered lithography limitations or high-aspect-ratio manufacturability requirements. Finally, following recent advances in diffusion models, our method can be further enhanced in several directions. One direction is projection-based guidance, where Zirvi et al.⁴⁸ use Singular Value Decomposition (SVD) of intermediate diffusion states to mitigate overly strong measurement gradients. Another is the adjoint formulation proposed by Blasingame and Liu,⁴⁹ which enables memory-efficient gradient computation during diffusion sampling. Both approaches assume smooth, continuous optimization variables and therefore do not directly handle fabrication-aware constraints, which are inherently discrete and robustness-critical. For this reason, while the two approaches differ from our setting, they are complementary to our framework. A promising direction for future work is to integrate such continuous optimization strategies with fabrication-aware design to combine their respective advantages. A third direction is the use of latent diffusion models,⁵⁰ which enable high-resolution generation (up to 2048×2048) by operating in a compressed latent space.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsphotonics.5c00993>.

Introduction to background algorithms and additional results supporting our method (PDF)

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Notes

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We open our source code and simulation file freely accessible at <https://github.com/dongjin-seo2020/AdjointDiffusion>.

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